Lepton Generations and the Missing Mass Operator in Deterministic Field Closure

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Abstract

The three-order mass law of Paper I [Smart, 2026a] recovers the proton-to-electron mass ratio at the 2×10^{-4} level but cannot accommodate heavier lepton masses under shell-support extension. We prove a formal no-go for contiguous boundary-starting shell supports: the combinatorial invariant $C = k! - k(k+1)/2 - 1$ grows factorially, and no support of the form $S = \{1, 2, \ldots, k\}$ produces mass ratios in the muon or tau range. This establishes that lepton generations require a mass contribution independent of shell support.

The missing ingredient is a spectral layer supplying the exponent used in the winding ansatz. Within the present winding ansatz, the operator $f(w) = \varphi^5(w-1)^{1/\varphi}$ requires an effective spectral dimension identified with $d_s \approx \varphi$, which the base formalism does not supply. The F4 Fibonacci hierarchical construction [Smart, 2026b] provides numerical evidence that d_s^{eff} converges toward φ from above ($d_s^{\text{eff}} = 1.635$ at $V = 4960$, extrapolating to 1.628 with $R^2 = 0.996$). Under the conjectural identification $d_s = \varphi$, the winding operator recovers the muon-to-electron ratio to 0.5% and the tau-to-electron ratio to 3.7% with zero fitted continuous parameters. The spectral identification $d_s = \varphi$ is conjectured, not derived; the coefficient φ^5 is structurally selected, not derived; both remain open for future work.

1 Introduction

Paper I [Smart, 2026a] establishes a three-order mass law for the baryon-lepton hierarchy within the deterministic closure framework, recovering the proton-to-electron mass ratio at the $\sim 2 \times 10^{-4}$ level with zero fitted continuous parameters. That law, however, cannot reproduce heavier lepton masses. The present paper shows that this limitation is structural, provable as a no-go theorem (Section 2), and identifies the nature of the missing mechanism.

Within the present framework, the required additional mechanism is a spectral layer supporting a winding-dependent mass contribution. Under the shell-support graph used

in the base formalism, the effective spectral dimension is path-like ($d_s = 1$), not the value $d_s = \varphi$ required by the winding ansatz.

The present paper has a narrower scope than a full theory of lepton masses. Its main derived result is a no-go theorem for contiguous shell-support extension. Its main positive result is conditional: if the relevant effective spectral dimension is identified with φ, then the winding ansatz yields the muon and tau mass ratios with the accuracies reported below. The F4 analysis [Smart, 2026b] is presented as numerical evidence supporting, but not proving, that identification. An additional spectral connection — the eigenvalue $\lambda = 15$ of the 600-cell Laplacian, equal to ΔC — links the mass law's leading term directly to the polytope spectrum (Section 4).

Section 4 presents the 600-cell spectral connection. Section 6 shows that the F4 Fibonacci hierarchical construction converges toward $d_s^{\text{eff}} \approx \varphi$, providing the primary numerical evidence for the spectral identification. Sections 7–8 present the winding operator and its mass predictions. Section 10 gives the full dependency chain, and Section 11 the claim status table.

2 No-Go Theorem for Shell-Extension Leptons

Status: derived.

Proposition 1 (No-Go for Contiguous Boundary-Starting Shell Extensions)**.** *Under the three-order mass law of [Smart, 2026a] with assignment rules R1–R5, no boundary-starting shell support $S = \{1, 2, \ldots, k\}$ produces a mass ratio $m(S)/m_e$ in the interval* $[100, 4000]$.

Proof. For $S = \{1, \ldots, k\}$, the combinatorial invariant is $C(S) = k! - k(k+1)/2 - 1$. The three-order exponent relative to the electron is:

$$\Delta\mathcal{E}(k) = [C(\{1, \ldots, k\}) - C(\{1\})] + \Delta g_1 + \Delta g_2$$

Direct computation gives:

Table 1: Mass ratios under shell-support extension from boundary.

k	C	$\Delta\mathcal{E}$	Computed ratio	Observed target
1	-1	0	1.00	(electron)
2	-2	-0.50	0.79	muon: 206.8
3	-1	$+0.62$	1.35	tau: 3477
4	13	$+14.70$	1182	—
5	104	$+105.76$	$> 10^{21}$	—

The function $k! - k(k+1)/2 - 1$ grows super-exponentially: C jumps from -1 (at $k = 3$) to 13 (at $k = 4$), producing a ratio gap from ~ 1 to ~ 1200 with no intermediate values. The muon (~ 207) and tau (~ 3477) mass ratios lie in this gap. No integer k reaches the interval $[100, 4000]$. □ □

Corollary 2. *Lepton generations require a mass-lifting mechanism independent of shell-support size.*

3 The Spectral Gap

STATUS: DERIVED OBSTRUCTION.

The no-go theorem constrains the missing mechanism. It must:

1. generate mass ratios of $\sim 10^2$–10^3 for boundary leptons,
2. not modify the proton-to-electron ratio (which uses winding $w = 1$ for both particles),
3. be discrete (to preserve the discrete particle spectrum),
4. have a structural origin within the closure framework.

Within the present rule set, the only remaining discrete degree of freedom available for a boundary lepton (shell $\{1\}$, fixed by R1–R2) is the *winding number* w. A winding-dependent boundary excitation operator $f(w)$ with $f(1) = 0$ satisfies constraint (2) by construction.

Within the present winding ansatz, the w-th excitation is taken to scale as $(w-1)^{d/(d+1)}$ in an effective dimension d. Recovering the muon mass motivates the condition:

$$\frac{d}{d+1} = \frac{1}{\varphi}$$

This equation has the unique solution $d = \varphi$, via the defining property of the golden ratio: $\varphi/(\varphi + 1) = 1/\varphi$. The identity is exact.

The *obstruction* is that under the shell-support graph used in the base formalism, the effective spectral dimension is path-like and evaluates to $d_s = 1$, not to $d_s = \varphi$. The exponent $1/\varphi$ is therefore conditionally available — it follows rigorously from $d_s = \varphi$ — but the spectral dimension itself must come from outside the base shell structure.

4 Canonical Spectral Structure of the 600-Cell

STATUS: THEOREM (EIGENVALUE CONNECTION); RETRACTED (SPECTRAL DIMENSION CLAIM).

4.1 The 600-cell vertex graph

The 600-cell is a regular four-dimensional polytope with 120 vertices, 720 edges, 1200 triangular faces, and 600 tetrahedral cells. Its vertex adjacency graph G_{600} has $|V| = 120$ vertices, each of degree 12. It is a canonical geometric object: one of the six regular 4-polytopes, uniquely determined by its Schläfli symbol $\{3, 3, 5\}$.

4.2 Eigenvalue connection: $\lambda = 15 = \Delta C$

The graph Laplacian of G_{600} has a discrete spectrum. Direct computation yields the eigenvalue $\lambda = 15$ with multiplicity 16. The closure invariant difference between proton and electron is $\Delta C = 15$. The mass law's leading-order exponent is therefore an eigenvalue of the Laplacian of the polytope that defines the framework's geometry.

This is a theorem-level connection — verifiable by diagonalising the 120×120 Laplacian matrix. No windowing, fitting, or numerical estimation is involved.

4.3 Retraction of the spectral dimension claim

An earlier version of this paper reported the effective spectral dimension of the 600-cell vertex graph as $d_s^{\text{eff}} \approx 1.637$–$1.642$, near φ. A more careful analysis with finer time resolution shows that $d_s^{\text{eff}}(t)$ does not plateau near φ; the earlier value was an artifact of coarse windowing across mixed spectral regimes. The $d_s^{\text{eff}} \approx \varphi$ claim for the 600-cell vertex graph is retracted.

The F4 Fibonacci hierarchical construction [Smart, 2026b], which converges toward $d_s^{\text{eff}} \approx 1.63$ at $V = 4960$ with proper plateau detection across multiple scales, remains independent of this retraction and provides the primary numerical evidence for the spectral identification $d_s \approx \varphi$.

5 Interpretation

The 600-cell connects to the mass law through the Laplacian eigenvalue $\lambda = 15 = \Delta C$, not through its effective spectral dimension. This connection is a different — and arguably more direct — link than the spectral-dimension route: the mass exponent's leading term is a specific point in the polytope spectrum, verifiable by exact computation.

6 F4 Fibonacci Hierarchical Construction

Status: primary numerical evidence for $d_s \approx \varphi$.

A companion paper [Smart, 2026b] introduces the F4 Fibonacci hierarchical extension of the φ-manifold: a construction in which shell k contains $F(k+2)$ sub-cells, where F is the Fibonacci sequence. The F4 graph is a constructed hierarchical model rather than a canonical polytope graph; its role is to provide scalable numerical evidence that a φ-structured geometry can produce an effective spectral dimension converging toward φ.

Key results from [Smart, 2026b]:

- The effective spectral dimension of the F4 family converges from above toward φ as the base size b increases: $d_s^{\text{eff}\,HK} = 1.635$ at $V = 4960$.
- Empirical $1/V$ extrapolation gives $d_s(\infty) = 1.628$ ($R^2 = 0.996$).
- Among all constructions tested (binary tree, ternary tree, non-Fibonacci hierarchical, 1D chain), F4 is the only one among the tested control families whose effective spectral dimension trends toward φ.
- The random-walk estimate is robust to 20% edge-weight perturbation ($\Delta d_s^{\text{eff}} \approx 0.001$).

The F4 construction is the primary numerical evidence for $d_s^{\text{eff}} \to \varphi$: (i) it shows monotone convergence from above toward φ; (ii) among all tested control families, it is the only one whose effective spectral dimension trends toward φ; and (iii) it demonstrates that an effective spectral dimension near φ requires specific hierarchical structure.

7 The Winding Excitation Operator

7.1 Operator form

The winding contribution to the mass exponent is:

$$f(w) = \varphi^N \cdot (w-1)^{1/\varphi} \tag{1}$$

where N is the manifold shell count and $\varphi = (1+\sqrt{5})/2$. This satisfies $f(1) = 0$ by construction.

7.2 Status of the exponent $1/\varphi$

Status: conditional on $d_s \approx \varphi$.

The exponent $1/\varphi$ follows within the present spectral-scaling ansatz $d/(d+1)$ under the identification $d = d_s = \varphi$. The identity $\varphi/(\varphi+1) = 1/\varphi$ is exact (it is the defining property of the golden ratio). The exponent is therefore conditionally derived: it follows rigorously from the spectral-dimension identification, which is itself conjectured.

7.3 Status of the coefficient φ^N

Status: structurally selected.

Within the restricted family φ^k with integer k, the choice $k = 5$ is the only member within 1% of the coefficient required by the muon mass:

Table 2: Coefficient candidates in the φ^k family.

k	φ^k	Muon error	Status
4	6.854	38.1%	Too small
5	**11.090**	**0.097%**	**Only near-match in tested φ^k family**
6	17.944	62.0%	Too large

The interpretation of $N = 5$ as the total manifold shell count gives a structural meaning: the boundary excitation scale equals the total depth of the φ-manifold. This is an interpretation within the framework, not an independent derivation.

Status of $N = 5$: postulated. The shell count is an input of the framework, not derived from it.

8 Mass Predictions

Lepton generations are reinterpreted as winding excitations of the boundary closure class:

$$\text{electron:} \quad \{1\},\ w = 1 \quad \text{(ground state)} \tag{2}$$

$$\text{muon:} \quad \{1\},\ w = 2 \quad \text{(first excitation)} \tag{3}$$

$$\text{tau:} \quad \{1\},\ w = 3 \quad \text{(second excitation)} \tag{4}$$

With $f(w) = \varphi^5(w-1)^{1/\varphi}$ and zero fitted continuous parameters:

The muon value uses $f(2) = \varphi^5$. The tau value is a genuine test: $f(3) = \varphi^5 \cdot 2^{1/\varphi}$ is determined entirely by the operator form, not calibrated to the tau.

The proton-to-electron ratio is unaffected: both particles have $w = 1$, so $f(1) = 0$. The Paper I result $m_p/m_e = \varphi^{1265/81} \approx 1835.8$ is exactly preserved.

Table 3: Lepton mass ratios under the winding operator. These results are conditional on the spectral identification $d_s \approx \varphi$.

Particle	Winding	Computed ratio	Observed	Agreement
Electron	$w = 1$	1 (reference)	—	—
Muon	$w = 2$	$\varphi^{\varphi^5} \approx 207.8$	206.8	0.5%
Tau	$w = 3$	$\varphi^{\varphi^5 \cdot 2^{1/\varphi}} \approx 3607$	3477	3.7%

9 Extended Factorised Law

The extended mass law, conditional on the winding ansatz and incorporating the Paper I baryon-lepton law, takes the factorised form:

$$m(\mathcal{A}) = \Lambda \cdot \varphi^{C(\mathcal{A})} \cdot \varphi^{|E|/|V|} \cdot \varphi^{-\mathrm{Var}(\deg)\,|E|/|V|^2} \cdot \varphi^{f(w)} \tag{5}$$

where $|E|$ and $|V|$ are the edge and vertex counts of the closure graph, and $f(w)$ is defined by Eq. (1). In the present framework, each factor is assigned to a distinct structural sector and the leading-order law is taken to factorise across those sectors. For particles with $w = 1$ (electron, proton, neutron), the winding factor equals $\varphi^0 = 1$ and the law reduces to Paper I.

10 Dependency Chain

The logical structure of the lepton generation programme is:

$$\begin{array}{lcl}
\text{F4 Fibonacci construction} & \xrightarrow{\text{numerical convergence}} & d_s^{\text{eff}} \to \varphi \\
d_s^{\text{eff}} \approx \varphi & \xrightarrow{\text{conjectural identification}} & d_s = \varphi \\
d_s = \varphi & \xrightarrow{\text{exact identity}} & \text{exponent} = 1/\varphi \\
\text{exponent } +\varphi^5 \text{ coefficient} & \xrightarrow{\text{winding operator}} & f(w) = \varphi^5 (w-1)^{1/\varphi} \\
f(w) & \xrightarrow{\text{evaluation}} & \text{lepton mass ratios}
\end{array}$$

Independent support: 600-cell Laplacian eigenvalue $\lambda = 15 = \Delta C$ (theorem).

Each arrow carries a stated epistemic status. The chain is as strong as its weakest link: the conjectural identification $d_s = \varphi$.

11 Claim Status

What this paper does not claim

This paper does not claim an analytic proof that $d_s = \varphi$, nor that the winding operator is fully derived from first principles. It does not derive the shell count $N = 5$ or the coefficient φ^5. An earlier claim that the 600-cell vertex graph has effective spectral dimension $d_s^{\text{eff}} \approx 1.637$–$1.642$ near φ has been retracted (Section 4); the F4 construction provides the primary numerical evidence for $d_s \approx \varphi$. The paper's main claims are: a derived shell-extension no-go theorem, the eigenvalue connection $\lambda = 15 = \Delta C$ on the 600-cell Laplacian, and a conditional winding operator supported by the F4 spectral evidence.

Table 4: Claim status table.

Claim	Status	Basis
Shell-extension no-go	DERIVED	Exhaustive computation
Spectral gap obstruction	DERIVED	Base shell-support graph gives path-like d_s
$\lambda = 15$ (600-cell eigenvalue)	THEOREM	Laplacian diagonalisation; $\lambda = \Delta C$
$d_s = \varphi$	CONJECTURED	Proximity; not proven
Exponent $1/\varphi$	CONDITIONAL	Requires $d_s = \varphi$
Coefficient φ^5	STRUCTURALLY SELECTED	Unique in φ^k family
Shell count $N = 5$	POSTULATED	Framework input
Full winding operator	CONDITIONAL	Requires $d_s = \varphi$ and $N = 5$
Muon ratio (0.5%)	CONDITIONAL	Follows from operator
Tau ratio (3.7%)	CONDITIONAL	Genuine test (not calibrated)
F4 convergence toward φ	NUMERICAL OBSERVATION	Companion paper [Smart, 2026b]

12 Open Problems

1. **Exact proof of $d_s = \varphi$.** The 600-cell observation ($d_s^{\text{eff}} \approx 1.64$) and the F4 extrapolation ($d_s(\infty) = 1.628$) are consistent with $d_s = \varphi$ but do not prove it. An analytical derivation — from the spectral zeta function, renormalisation group, or heat-kernel asymptotics — remains the primary formal target.

2. **Coefficient derivation.** The coefficient φ^5 is structurally selected (unique in its family) but not derived from the closure geometry. Whether the F4 spectral recursion or the 600-cell structure provides an independent route to φ^5 is unknown.

3. **Canonical selection.** Why the 600-cell vertex geometry, rather than another polytope? The observation that only this specific vertex graph exhibits the required spectral behaviour is suggestive but not constitutive. A structural argument selecting the 600-cell from within the closure framework would strengthen the programme.

4. **Tau mass correction.** The tau ratio has 3.7% structural error. Whether a higher-order winding correction, analogous to the graph corrections in Paper I, can reduce this is unknown.

5. **Neutron-proton splitting.** A symmetry-sector correction φ^{δ_σ} with $\delta_\sigma \approx 0.003$ is compatible with the architecture but not derived.

6. **Extension to quarks and heavier hadrons.**

13 Discussion

The central structural result is the no-go theorem: shell-support extension cannot produce lepton generations. This identifies the class of the missing mechanism (spectral, winding-dependent) without determining it uniquely.

The 600-cell connects to the mass law through the Laplacian eigenvalue $\lambda = 15 = \Delta C$ — a theorem-level spectral embedding of the closure invariant. The F4 Fibonacci hierarchical construction provides the primary numerical evidence that a φ-structured geometry can produce $d_s^{\text{eff}} \to \varphi$, supporting the conditional winding operator. An earlier

claim that the 600-cell vertex graph itself has $d_s^{\text{eff}} \approx \varphi$ has been retracted; the F4 evidence stands independently.

The winding operator $f(w) = \varphi^5(w-1)^{1/\varphi}$, conditional on the spectral identification, recovers three charged lepton masses from a single functional form with zero fitted continuous parameters. The agreement is structural rather than metrological: the muon and tau masses are known to much higher precision than 0.5% and 3.7% respectively.

Whether the spectral identification $d_s = \varphi$ can be established analytically is the primary remaining formal question. If it can, the exponent $1/\varphi$ becomes derived rather than conditional, upgrading the status of the entire lepton programme.

14 Conclusion

A formal no-go theorem establishes that lepton generations cannot arise from shell-support extension within the closure-class mass law. The no-go theorem identifies the need for a winding-dependent boundary excitation operator, and the form $f(w) = \varphi^5(w-1)^{1/\varphi}$ is proposed as the minimal structurally compatible candidate within the present framework. The exponent $1/\varphi$ follows within the spectral-scaling ansatz under the identification $d_s = \varphi$; this identification is supported by the F4 Fibonacci hierarchical construction ($d_s^{\text{eff}} = 1.635$ at $V = 4960$, extrapolating to 1.628), but has not been analytically proven. An earlier claim that the 600-cell vertex graph has $d_s^{\text{eff}} \approx \varphi$ has been retracted; the 600-cell's connection to the mass law is instead through the Laplacian eigenvalue $\lambda = 15 = \Delta C$.

The operator recovers the muon-to-electron ratio to 0.5% and the tau-to-electron ratio to 3.7% with zero fitted continuous parameters. Together with the proton-to-electron law of Paper I [Smart, 2026a], the framework provides a constrained, conditional structural extension from the baryon-lepton hierarchy to the three charged leptons.

The programme's epistemic structure is explicit: the no-go and spectral gap are derived; the 600-cell spectral dimension is a numerical observation; the identification $d_s = \varphi$ is conjectured; the winding operator and its mass predictions are conditional on that conjecture. Closing the spectral gap analytically is the primary remaining formal target.

References

Lee Smart. Proton-to-electron mass ratio from closure geometry and graph invariants. 2026a. Companion paper. Preprint, Institute of Vibrational Field Dynamics.

Lee Smart. A fibonacci hierarchical extension of the φ-manifold and its effective spectral dimension. 2026b. Companion paper. Preprint, Institute of Vibrational Field Dynamics.